Essays

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Discuss, with examples, how biotechnology contributes to different aspects of medicine. Briefly explain the working principles of the techniques involved.

(12 marks)

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Diagnosis of genetic diseases (Max.4)

- DNA fingerprinting is used to diagnose genetic diseases in both prenatal and newborn babies. These diseases may include cystic fibrosis, haemophilia, Huntington's disease, Alzheimer's disease, sickle-cell anaemia and thalassemia. (1)
- DNA fingerprinting is a technique for finding unique patterns in a person's DNA. (1) By comparing large groups of people with and without a particular genetic disease, it is possible to identify DNA patterns associated with the disease. (1)
- DNA sequencing is a method used to determine the order of the bases in DNA. (1) DNA sequencing can be used to identify gene mutations. (1)
- Gene mapping means locating genes in the genome. (1) This technique allows the identification of genes that are related to diseases, which can lead to the development of new diagnostic tests. (1)

Treatments (Max. 3)

- Bacteria can be genetically modified to produce pharmaceutical products (e.g. insulin, vaccines). (1)
- Recombinant DNA technology allows human genes to be transferred to bacteria and expressed. (1)
- The gene products are expected to be identical to those found in humans, lowering the risk of immune responses and possibly increasing the effectiveness of the treatment. (1)
- In gene therapy, genes are transferred into a patient's cells to treat diseases. (1)
- Data from the human genome project / knowledge of the human genome helps the development of personalized medicines. (1)

Prevention of diseases (Max. 2)

- Data from the human genome project / knowledge of the human genome helps scientists better understand how genes are inherited and how genes mutate. (1)
- Understanding the human genome enable scientists to assess risks posed to individuals by exposure to toxic agents / carcinogens, especially in terms of cancer risk. (1)
- Accurate risk assessment would lead to better prevention of diseases. (1)

Communication (Max. 3)

(12 marks)

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Skills assessed	Marks
Applying concepts	9
Communication	3

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